WEATHER ALERT LAB

Lab 7 — Fuzz Testing and Input Validation

This report records the initial test setup, application execution, and the final successful property-based test run. The screenshots are the execution evidence.

1. Initial run

The first test-file attempt did not collect tests because the three starter test functions had no indented bodies. Pytest reported an IndentationError. The application was also run with sample temperature, wind, and rainfall inputs.

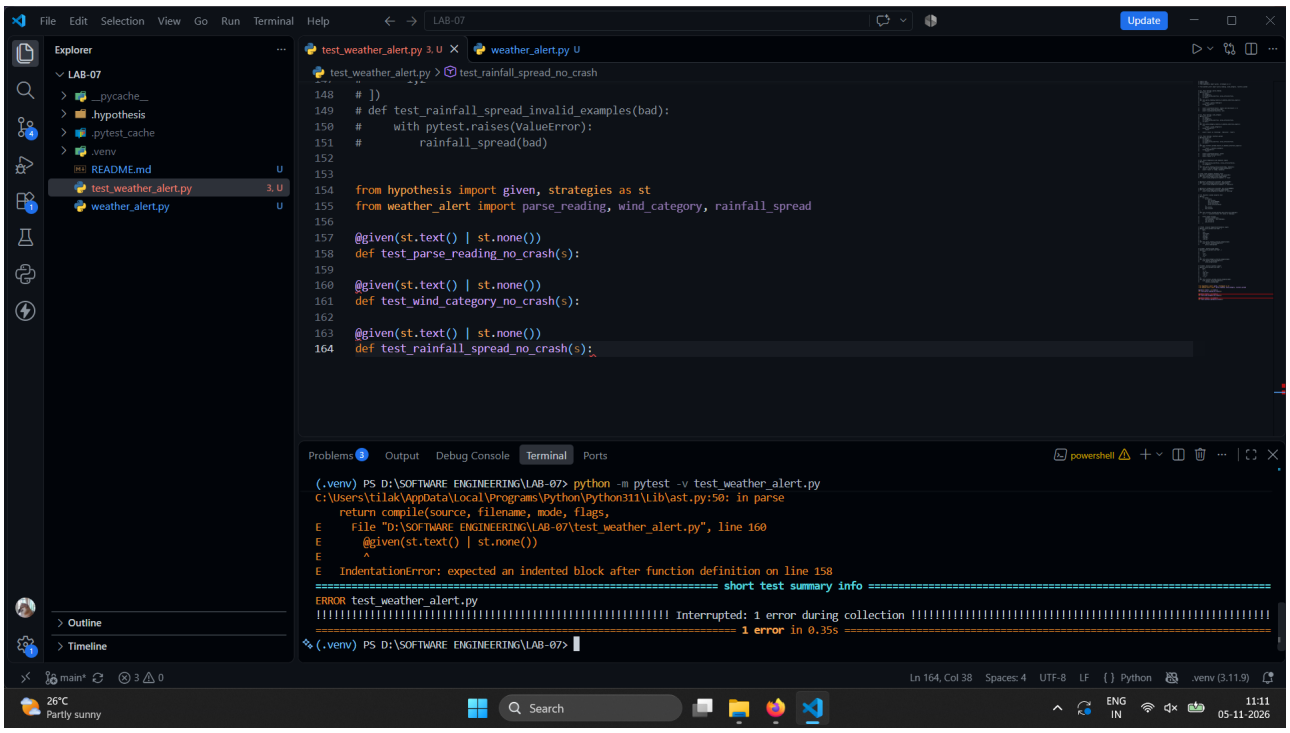


Figure 1. Initial test attempt — collection stopped due to incomplete test functions.

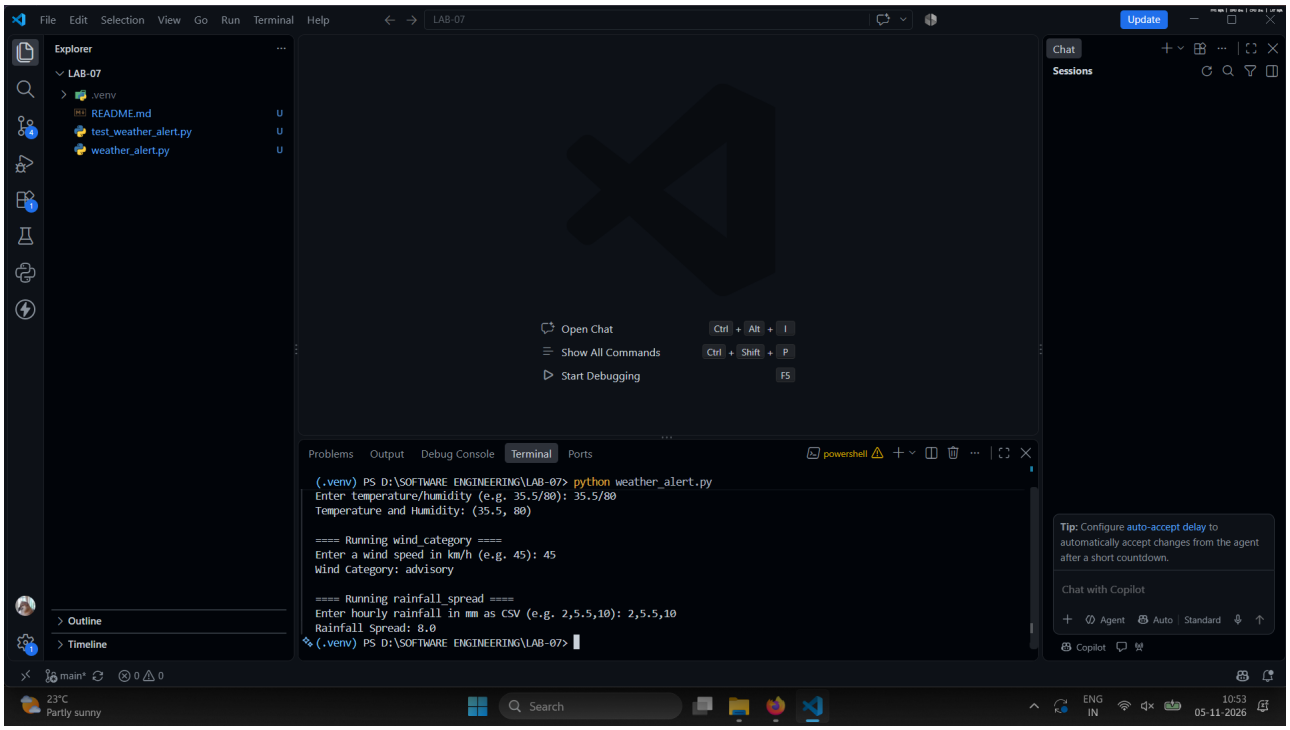


Figure 2. Weather alert application run with sample inputs.

2. Updated tests and final run

The test file was completed with Hypothesis-based input generation, valid-input property checks, boundary checks for wind categories, and invalid-input examples for all three functions.

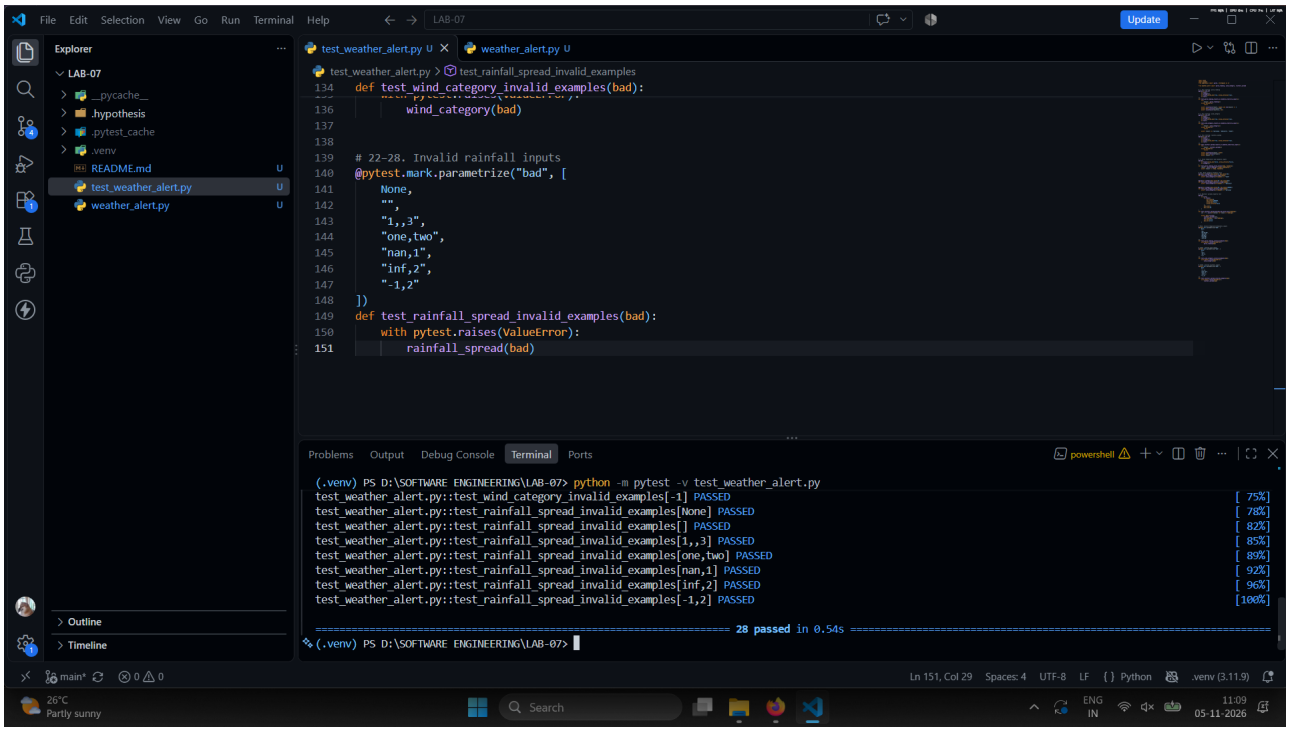


Figure 3. Final pytest run — all 28 test cases passed.

The application was then executed with sample values: temperature/humidity 35.5/80, wind speed 45 km/h, and rainfall readings 2, 5.5, 10 mm. The displayed results were (35.5, 80), advisory, and rainfall spread 8.0 mm.

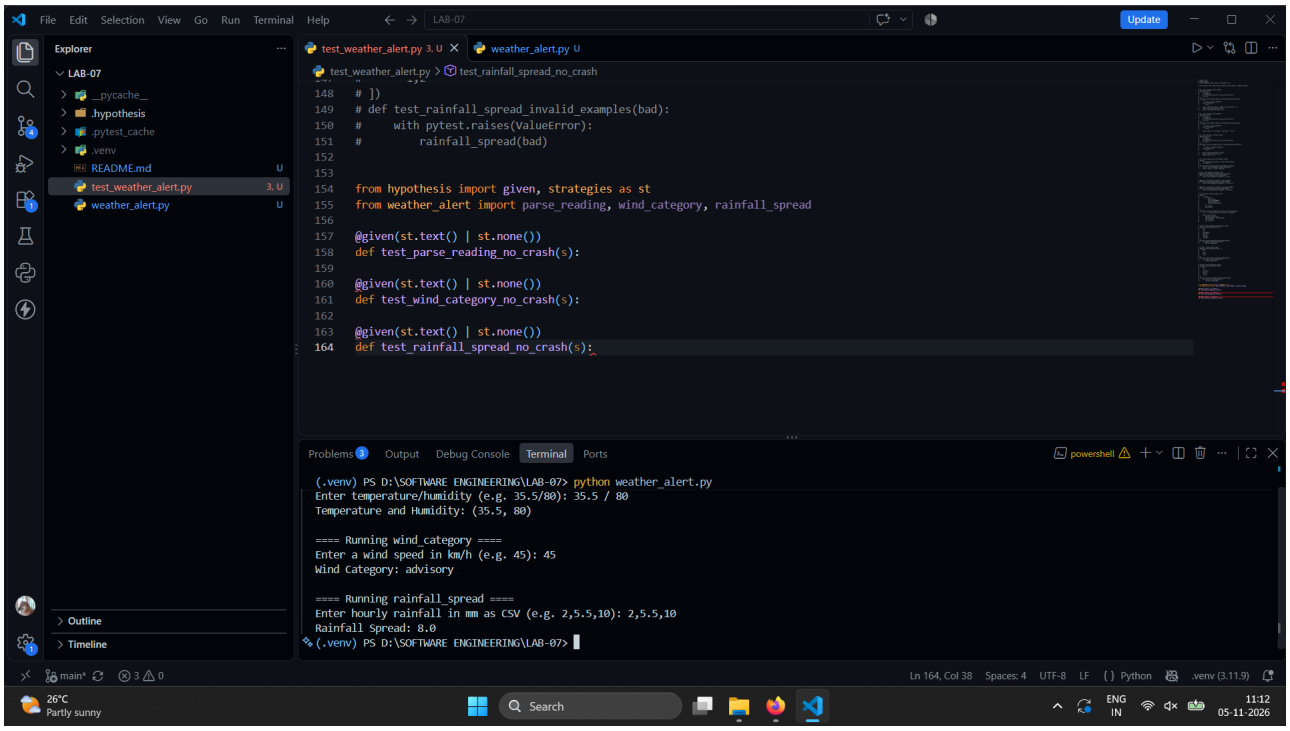


Figure 4. Application execution after the test and validation updates.

Final result: 28 tests passed. The test suite covers arbitrary inputs, valid values, category boundaries, and representative invalid inputs.